

DEFINING OF RELIABILITY

The term reliability is defined as the probability that a component, item or assembly will operate without failure for a specific period of time under specified conditions in order to ensure an meaningful results conditions of operations both physical and electrical conditions that must be specified in detail as follows.

THE ABOVE CONDITIONS INCLUDE:-

1. Air pressure
2. External conditions such as exposure to sand and dust storms, rain storm, salt water spray or solar radiation.
3. Maximum and minimum humidity
4. Maximum and minimum ambient temperature.
5. Shielding from ionizing radiation which they may experience.

ELECTRICAL CONDITIONS

1. Variation in power supply voltage and size of the voltage transients.
2. Variation in loading where relevant.
3. Variation in the shock level.
4. With AC supply the variation in frequency and harmonic contents.
5. The level of unwanted RF energy radiated by the equipment must not cause interference to radio communications.
6. The equipment must be able to tolerate some RF radiation if it is to be used near high power or radio transformers.

Based on some mathematical reasons reliability is considered to be 0 and 1 eg.

Reliability = 1.0 $\Rightarrow$ This means the item will operate always as indicated by the manufacturer.

Reliability 0.90 $\Rightarrow$ It 90% likely to work as work as indicated.

Reliability 0.00 $\Rightarrow$ Means absolutely it will not work as indicated.

IMPORTANCE OF RELIABILITY

1. Complexity of Equipment | System: Because of persistent demands of equipments from military, communication, and aerospace sector for example. There has been a growing need to develop complex equipment | system upon which strikes requirements for very high reliability to be put in place.

It should be noted that some equipment | system are complex in nature due to their multivarious features or multivariied functions. They are expected to perform

This therefore calls for a very high reliability on such equipment | system. eg. A Space Craft launched into the space is expected to remain in their working condition for several years without being brought down repairs because it has been built to high reliability not self

repair facilities under this condition, it is obvious that achievement of high reliability of system | equipment is synonymous with increase complexity of design and construction.

b) Cost of Repairs And Maintenance:- Reliability is also important with respect to cost of repairs and maintenance. Typical examples of all electronic equipment used by the Nigerian Armed forces in the Biafra war is inoperative at all times due to environmental problems as a result the maintenance of individual electronic equipment consumed up to ten times the initial cost price over a certain period of time. This is because the equipment were not built with very high reliability components that can withstand the test of our environment eg. Repeaters which are used in water communication system which must operate without failure for a very long time. In this case the very high cost of repairs and maintenance and cost of the downtime debt the need for high reliability.

c) Perceived Risk: Market competition, rapid evolution of new materials and methods, the need to reduce

(2)

Product cost, Safety and legal consideration, warranty cost etc constitute to overall perception of risk.

Reliability study therefore has developed in response to the need to combat these risk eg. Manufacturers often suffer high cost for the failure of products under warranty outside the warranty period only the customers suffers. In any case the manufacturers of product with low reliability may also incur a loss of reputation and subsequent effects in his/her business prospect. Hence reliability is important.

FAILURE

- 1) An item is considered to have failed when it undergoes the or satisfies the following conditions
- 2) When it becomes completely inoperative.
- 3) Operative but unable to perform its required functions eg. An electronic motor now rotates at a speed lower than normal.

Unsafe for its continues use eg. An electronic item produces a shock on the user.
Failure of a component can therefore be defined as the inability of the item or component to accomplish its intended purpose. A fault is there or least to as an imperfection in an item or machine or component. An item may be referred to as a distinct part that can be specified separately in a group of things that could be enumerated on a list.

CLASSIFICATION OF FAILURE

- 1) Failures may be classified according to:
 - a) Causes of failure
 - b) Timing of failure
 - c) Degree of failure
 - d) Combination of timing causes and degree of the fault.
- 2) Cause of Failure:- There are two types failure.

a) Misuse fault :- This failure is attributed to an item when subjected to stresses beyond the capability of its performance.

b) Inherent weakness. This is failure that is attributed to an item when subjected to inherent weakness within the stated stresses of its capability of its performance.

Combinational fault :-

(i) Catastrophic failure :- This is a failure that is sudden and complete eg. open circuit in the wire wound resistor.

(ii) Degradational failure :- This is failure that is both gradual and partial when subjected to stresses eg. the fading away of colour coded capacitor.

Degree of failure :- This is failure that have either partial failure which is attributed to an item when it undergoes gradual stages of stresses or begins to reduce in quality due to wear out part of the item itself.

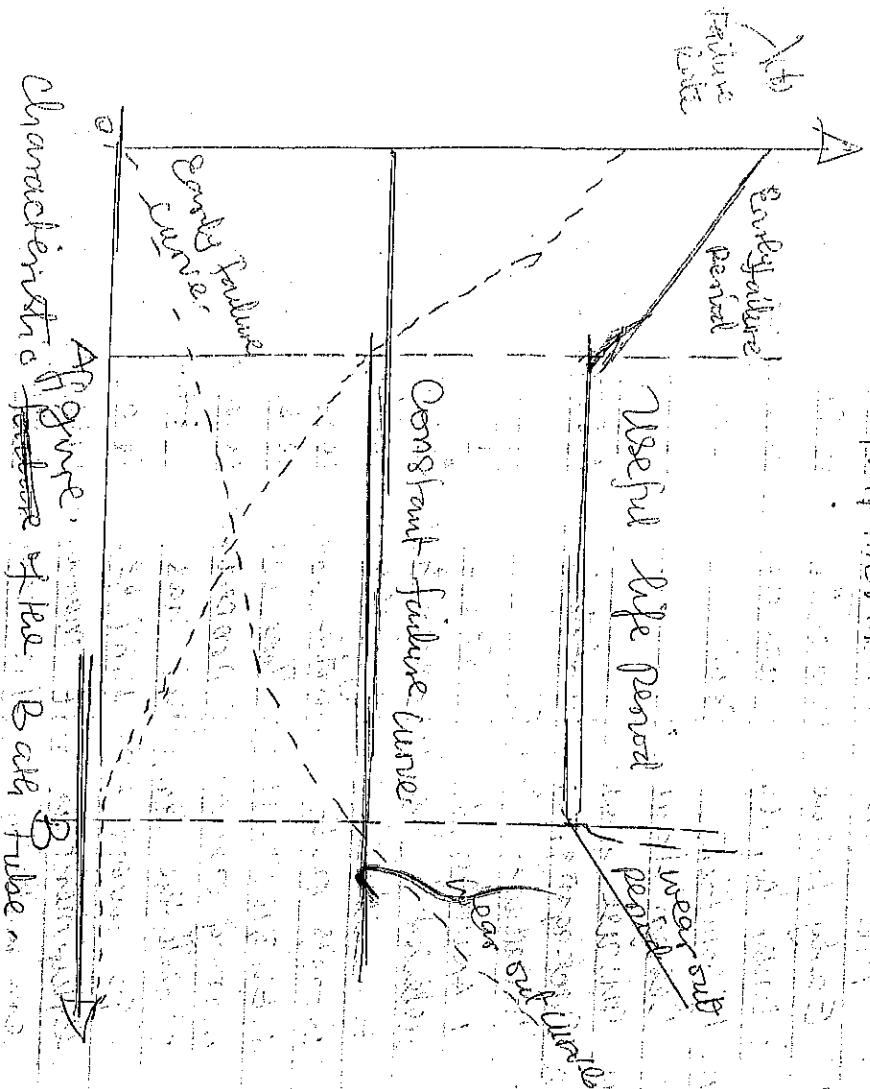
Failure Due to Timing :- There are two types of failures due to timing.

(i) Gradual failure :- This failure that is attributed to product examination.

(ii) Complete failure :- This failure type is attributed after examination.

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(2) FAILURE PATTERN: (THE BATH TUB CURVE) OF AN EQUIPMENT



Characteristics of the Bath Tub Curve

The bath tub curve is obtained from the addition of three curves which are:

- EARLY FAILURE CURVE:** Having an exponential decreasing pattern which reflects failure rate due to weak parts that escape final testing or screening but, when subjected to normal operational stress succumb to infant mortality.

- RANDOM FAILURE CONSTANT CURVE:** Having constant or straight line which reflects relatively low failure rate would occur if no weak parts inherent wear out mechanism were present into clearance failure.

- THE WEAR OUT CURVE:** Showing an exponential increasing pattern due mainly when some component in an equipment begins to exhibit an inherent wear out mechanism.

CAUSES OF FAILURE IN EACH OF THE REGION OF THE BATH TUBE

1. Early failure period :- The most common of early failure under the region are due to the following.

(i) Manufacturing fault.

(ii) Design faults

(iii) Misuse faults

(iv) Packaging, transportation and storage faults

(v) Installation faults.

MANUFACTURERS FAULTS:- These are those which were not detected before the design of an item to the consumer. These types of fault are usually associated with joints, connections, welds, cuts, impurities and cracks.

DESIGN FAULT:- These are those kinds of faults caused by wrong designing procedures which may be undetected until the equipment has been sold out to the customers.

To avoid this kind of faults a prototype of an item or equipment is first manufactured and properly tested before embarking on the main production.

MISUSE FAULTS:- This kind of fault could occur on an item or equipment due to incompetent operation of the equipment or item most probably by a customer or any body else. It may also be due operating an item in hazardous environment for which it was not designed.

PACKAGING AND TRANSPORTATION STORAGE FAULTS

The major factor which may be responsible for failure during packaging and transportation as well as storage includes the following:

a) Inadequate or suitable packaging materials for the type of exportation used.

b) Inadequate Marketing :- Pictorial handling instructions such as marking as:- Carrying in vertical position only should be indicated on the outside the case containing the items. This include drawing of umbrella which is often used.

c) Degradation of Packaging material due to methods of storage prior to use.

V) INSTALLATION FAULTS:-

Incorrect or poor installation of an item or equipment may also cause an incorrect or poor installation of equipment. There are specific precautions to be taken in order to avoid equipment early failure.

VI) CONSTANT FAULTS

As soon as the early failure period is over other parts of the equipment or system settles down for a reasonable long period of time during which the equipment or system is expected to have a stress related to random (fluctuation) of stress excluding the component strength. This is the reason why this period is ~~also~~ referred to as random failure period or stress related period.

When the failure period is constant in this period reliability stress related period.

When the failure period is constant as in this period reliability production is made more easy. Mathematically because the expression $R = e^{-\lambda t}$ if earlier can be readily used for reliability assessment. Below are the examples of system and their approximate useful life period.

System

Greater life period.

Telephone

Greater than 15 years

Electric Power System

Greater than 15 years

Television Set

Seven to 10 years based

on 200 hrs per year.

Air traffic Control

Greater than 15 years.

System

WEAR OUT FAILURE PERIOD

When an equipment system comes to the end of its useful life period, its failure rate may increase because in addition to chance failures parts starts to deteriorate and wear out. This may be caused by corrosion oxidation breakdown of installation fluctuation wear

EEEC 324 NOTES

LECTURES COPY

CLASSIFICATION OF FAILURES.

1. Causes of failure: There are two types of fault in this category.

- a) Misuse failure: This type of failure is attributed to an item or system when they are subjected to stresses beyond the capability of ~~the~~ ^{their} performance. Inherent weakness. This kind of failure of an equipment is attributed to ^{or} ~~subjection~~ ^{subjection} by inherent weakness within the stated stress of its capability of its performance.

2. Combinational Failure: - There are two types of Combinational failure categories viz a) Catastrophic Failure: - This is a failure that is sudden and complete eg open circuit in a wire wound resistor.

- b) Degradational Failure This type of failure is both gradual and Partial when subjected to stress eg. The fading away of color coded capacitor Degree of failure: - Under this we have only Partial failure which is attributed to an item when it undergoes gradual stages of stresses and begins to reduce in quality due to worn out parts of the item itself.

4. Failure Due to Timing: - The two types of failure due to timing are explained as follows:

- 1) Gradual failure: This type of failure of an item is attributed prior examination.
- b) Complete failure: This is failure that is not attributed to prior examination.

TYPICAL VALUES OF FAILURE RATES OF SOME COMPONENTS:

1. Capacitor (Paper film type)	0.1
2. Resistor (Carbon film type)	0.05
3. Diode (Silicon 1 watt)	0.005
4. Transistors	0.3
5. Triode Valve	1.8
6. Transistor (RF type)	0.6

The Organization where these types of informations are gotten from are

- (i) The Standard Part report dealers on these products
- (ii) These reports can also be gotten from exporters
- (iii) Reports also from the manufacturers.
- (iv) Report from the users.
- (v) Maintenance and repairs reports.

Component failure rate usually are affected by the mechanical, electrical and thermal environment required for operations. For this reason the effect of environmental temperature and the operational stress on the failure rate of each component is accounted for by introducing multiplying coefficient wt, wt is known as identifying factors due to environment temperature and rating respectively.

FAILURE RATE OF COMPONENTS

The failure rate of an item or component is the total number of times an item or component will not accomplish their intended purpose during operation per hour. The overall failure rate of each component of an equipment can be obtained as follows

$$A_0 = A_{WE} \cdot W_T \cdot W_R \quad \text{--- (1)}$$

where A = Basic failure

~~total~~ W_T = weighting factor due to environment

W_R = weighting factor due to Rating

W_T = weighting factor due to temperature

MEAN TIME BETWEEN FAILURES:

This is defined as the mean value of length of time which elapses between failure. The are two basic methods to calculate the mean time between failure of component. The two test methods are

①
②

1. Non replacement method

2. Replace ment Method: The MTBF (M) may be measured by testing for a total period of time t_n where

t_0 = Time that the test started

t_n = time that the number of n failure during which, n fault or each fault required and put back to use.

T = Result in the observed MTBF given by

$$M = T/n \Rightarrow M = (t_1 - t_0) + (t_2 - t_1) + \dots + (t_n - t_{n-1})$$

n

②

If $t_0 = 0$ is the reference starting point or time where

n = Number of failures

t_0 = time of start of test

t_1 = time of first fail

t_2 = time of 2nd fail

t_n = time to n^{th} failure

Example ①

Components were conducted on an electronic items and faults which were reported. Modelledy occurred at the following times.

Table ①

Failure	0	1	1	1	1	1	1
Time (X hrs)	$t_0 = 0$	$t_1 = 2$	$t_2 = 4$	$t_3 = 8$	$t_4 = 10$	$t_5 = 14$	$t_6 = 15$

Calculate the MTBF

Solution:-

using eq ②

$$MTBF = \frac{M}{n} = \frac{t_n - t_0}{n}$$

$$\frac{15 - 0 \times 10^{+2}}{C_0} = \frac{15 \times 100}{6} = \underline{\underline{250 \text{ hrs.}}}$$

2. Non replacement method: This requires that a large number of the items be put under test and observations to be carried out for possible failure at the beginning and the end of the period. In this case the formula for MTBF $M = \text{test hours for failures} + \text{test hours for survivors}$.

$$\text{i.e. } M = \text{Total component test hours survivors} \dots \text{eq (8)}$$

Example (2)

In order to determine the MTBF of a certain components 50 were tested for a period lasting for 200 hrs. The time to failure of the components are shown in the table below. 35 components survived without failure. Assuming the wear out failure ignored -

Calculate

- (i) Total test hours before failure
- (ii) Total test hours without failure
- (iii) Total ~~test~~ survival hours
- IV MTBF.

Solution

Table (2)

N ^o of Components	Time to failure in hours
6	100
5	140
4	175
Total test hours before failure	
= $(6 \times 100) + (5 \times 140) + (4 \times 175)$	
= $600 + 700 + 700$	
= <u><u>2000 hrs</u></u>	

(i) Total test hours without failure
 $= 35 \times 200 = 7000 \text{ hrs}$

(ii) Total Survival hrs
 $= (2000 + 7000) \text{ hrs}$
 $= 9000 \text{ hrs}$

MTBF = $\frac{N}{\text{Total Survival hours}}$
 $= \frac{9000}{15} \text{ hrs}$
 $= 600 \text{ hrs}$

MTTF (Mean Time To Failure)

This is defined as the average total time an item may be expected to function before failure. This estimate can be computed after testing for number of items N using specified way by applying electrical and mechanical heat or humidity conditions until all items have failed. If the item to failure are $t_1, t_2, t_3, \dots, t_n$ then we observed MTTF is given by

$$\text{MTTF} = \frac{1}{N} \sum_{t=1}^n (t_i - t_0)$$

Thus $\text{MTTF} = \frac{1}{N} [(t_1 - t_0) + (t_2 - t_0) + \dots + (t_n - t_0)]$
 where t_0 is equal to starting time or reference
 $(t_1 - t_0)$ - Period to first failure
 $(t_2 - t_0)$ - Period to second failure
 $(t_n - t_0)$ - Period to N^{th} failure.

N = Total number of failure of components
 Example 2

$$\frac{15 \times 10^4}{C} = \frac{15 \times 100}{6} = 250 \text{ hrs.}$$

2. Non replacement Method: This requires that a large number of the items be put into under test and observations to be carried out for possible failure at the beginning and the end of the period. In this case the formula for MTR, $M = \text{test hours for failures} + \text{test hours for survivors}$.

$$M = \text{Total Component test hours survivors} + \text{Total Component test hours failures} \quad \text{--- eq (2)}$$

Example (2)

In order to determine the MTR of a certain component 50 were tested for a period lasting for 200 hrs. The time to failure of the components are shown in the table below. 35 components survived without failure. Assuming the wear out failure ignored -

Calculate

- (i) Total test hours before failure
- (ii) Total test hours without failure
- (iii) Total test survival hours
- IV MTR.

Solution

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= <u>2000 hrs</u>	

(ii) Total test hours without failure
 $= 35 \times 200 = 7000 \text{ hrs}$

(iv) Total Survival hrs.
 $= (2000 + 7000) \text{ hrs}$
 $= 9000 \text{ hrs}$

MTBF = $N = \frac{\text{Total Survival hours}}{\text{Number of Components}}$

$= \left(\frac{9000}{15} \right) \text{ hrs}$
 $= 600 \text{ hrs}$

MTTF (Mean Time To Failure)

This is defined as the average total time an item may be expected to function before failure.

This estimate can be computed after testing for number of items N in a specified way by applying electrical and mechanical heat or humidity conditions until all items have failed. If the item to failure are $t_1, t_2, t_3, \dots, t_n$ then we observed MTTF is given by

$MTTF = \sum_{t=1}^n (t_i - t_0)$

Thus $MTTF = (t_1 - t_0) + (t_2 - t_0) + \dots + (t_n - t_0)$ where t_0 is equal to starting time or reference

$(t_1 - t_0)$ - Period to first failure

$(t_2 - t_0)$ - Period to second failure

$(t_n - t_0)$ - Period to N^{th} failure.

$N = \text{Total number of failure of components}$

Example 2

Life testing made on six non repairable electrical lamps and the following result were obtained -

Failure	0	1	1	1	1	1
Time $\times 100^2$	$t_1 = 4$	$t_2 = 10$	$t_3 = 16$	$t_4 = 20$	$t_5 = 23$	

Calculate the M.T.T.F of the component.

from eq. (3)

$$M.T.T.F = (t_1 - t_0) + (t_2 - t_1) + (t_3 - t_2) + (t_4 - t_3) + (t_5 - t_4)$$

$$= 4 - 0 + 10 - 4 + 16 - 10 + 20 - 16 + 23 - 20$$

$$= 4 + 10 + 16 + 20 + 23$$

$$= \frac{73 \times 10^2}{6}$$

$$= \frac{73 \times 100}{6} = 1216.6 \text{ hrs}$$

RELIABILITY AND UNRELIABILITY

if failure rate is constant the probability of

no failure occurring in a given time is

$$R = e^{-\lambda t} \text{ where } R \text{ is the probability of no}$$

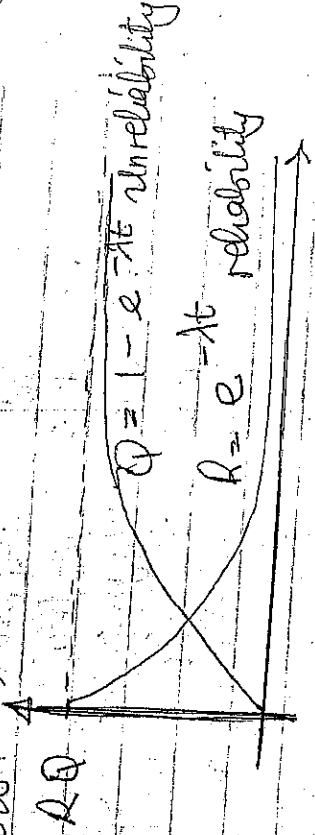
failure in time t . i.e. reliability. The unreliability

Q is defined as the probability of product failure. It follows from statistical theory that

$$R + Q = 1$$

$$Q = 1 - R = 1 - e^{-\lambda t} \text{ eq (4) and that}$$

Graph of R & Q against time exponential as shown below:



MATHEMATICAL M.T.T.R. RELATIONSHIP OR TERM

We assume failure rate of the graph to be constant and derive mathematical expression which relate the following

i) Reliability and failure rate is $R = e^{-\lambda t}$

ii) M.T.B.F, M.T.P and failure rate is $M = \lambda \times \text{Taking}$

from (i) reliability and failure rate is $R = e^{-\lambda t}$
Consider the case in which a fixed number of identical components are tested.

Let N_S = Number of surviving upto time t

N_f = Number of surviving failed to time t

$N_0 = N_S + N_f$ = Total number of operation at time $t = 0$

$$R(t) = \frac{N_S}{N_0}$$

from eq (i) N_0 = reliability at any time t .

$$1 = \frac{1}{N_0} \times \frac{dN_f}{dt} \Rightarrow 1 = \left(\frac{1}{N_0 - N_f} \right) \times \frac{dN_f}{dt}$$

$$\text{or } \int_0^{N_f} \frac{dN_f}{N_0 - N_f} = -\lambda t \left[\log_e (N_0 - N_f) \right]_0^{N_f}$$

$$\lambda t = \left[\log_e (N_0 - N_f) \right]_0^{N_f}$$

$$\Rightarrow -\lambda t = \left[\log_e (N_0 - N_f) - \log_e (N_0 - 0) \right]$$

$$\Rightarrow -\lambda t = \log_e (N_0 - N_f)$$

$$\text{Thus } e^{-\lambda t} = \left(\frac{N_0 - N_f}{N_0} \right) \quad \#$$

$$\text{Rout } R(t) = \frac{N_S}{N_0} = \frac{N_0 - N_f}{N_0} = 1 - \frac{N_f}{N_0}$$

$$R_t = 1 - e^{-\lambda t}$$

If we compare eq. * and ** we have $R_t = e^{-\lambda t}$
 If the rate is not constant as previously assume
 then

$$R(t) = e^{-\int_0^t \lambda(t) dt} \quad \text{--- eq (6)}$$

(ii) MTBF, MTTF and failure rate = $\lambda_m = 1$
 Note that the general expression for MTBF λ
 is $M = \int_0^\infty R(t) dt$ --- eq (7)

$$\int_0^\infty R(t) dt \quad \text{--- eq (7)}$$

For the case when λ is constant $R = e^{-\lambda t}$ and
 equation (7) becomes $M = \int_0^\infty e^{-\lambda t} dt = - \left[\frac{1}{\lambda} e^{-\lambda t} \right]_0^\infty$

$$\therefore \text{MTBF}, \left[M = \frac{1}{\lambda} \right] \quad \text{--- eq (8)}$$

SURVIVOR AND FAILURE EQUATION CURVES

From eq. * and (5) $R = 1 - \frac{N_f}{N_0} = e^{-\lambda t}$ --- (9)

therefore $N_s = N_0 - N_f = N_0 e^{-\lambda t}$ or
 $N_s = N_0 e^{-\lambda t}$ --- eq (10)

Equation (10) is the equation of the graph of
 survivor against time (survivor curve) similarly
 from eq (9) it can be easily shown that

$$N_f = N_0 (1 - e^{-\lambda t}) \quad \text{--- eq (11)}$$

$$\Rightarrow \frac{N_f}{N_0}$$

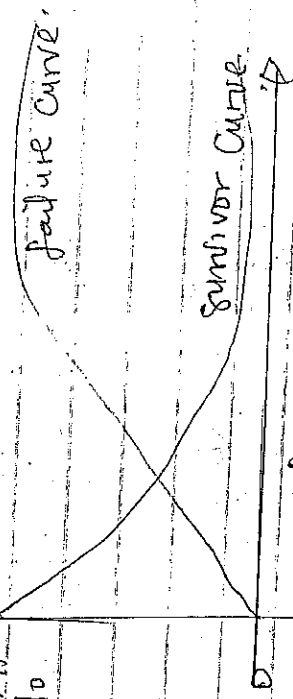


fig (2)

The fig (2) above shows survivor and failure curve
 from eq (1) $-R = \frac{N_s}{N_0} = e^{-\lambda t}$

Also from equation (9) $1 - \frac{N_f}{N_0} = (1 - e^{-\lambda t})$

Example ④

One thousand similar components each of constant failure rate of $5\%/10^3$ hrs are put to test together. Calculate the time lag before failure of the following number of the components.

100

500

Solution

$$\text{Recall } R = e^{-\lambda t} = 1 - \frac{NF}{N_0}$$

$$\text{But } \lambda = \frac{5 \times 1}{1000} = \frac{5}{1000}$$

$t = t_1 =$ time required before failure of 100 components.

$$N_0 = 1000, NF = 100$$

$$\text{Then the exponential } \left(-5 \times 10^{-5} t_1 \right) = 1 - \frac{100}{1000}$$

$$= 1 - 0.1$$

$$= -5 \times 10^{-5} t_1 = \log 0.9$$

$$\therefore t_1 = \underline{\underline{2107 \text{ hrs}}}$$

(ii) Let $t_2 =$ time required for 500 components to fail then

$$\text{exp. } \left(-5 \times 10^{-5} t_2 \right) = 1 - \frac{500}{1000} = 1 - 0.5$$

$$= -5 \times 10^{-5} t_2 = \log 0.5$$

$$\therefore t_2 = \underline{\underline{13863 \text{ hrs}}}$$

If we compare eq. * and * we have $R(t) = e^{-\lambda t}$
 If the rate is not constant as previously assumed
 then

$$R(t) = e^{-\lambda t} \quad \text{--- eq. (6)}$$

ii) MTF, MTFE and failure rate = λ
 Note that the general expression for $m(t)$ is $M = \int_0^t R(t) dt$ --- eq. (7)

For the case when λ is constant $R = e^{-\lambda t}$ and
 equation (7) becomes $M = \int_0^t e^{-\lambda t} dt = -\left[\frac{1}{\lambda} e^{-\lambda t} \right]_0^t$

$$\therefore \text{MTBE}, \left[M = \frac{1}{\lambda} \right]_0^t \quad \text{--- eq. (8)}$$

SURVIVOR AND FAILURE EQUATION CURVES

From eq. * and (3) $R = 1 - \frac{N_f}{N_0} = e^{-\lambda t}$ --- (9)
 therefore $N_s = N_0 - N_f = N_0 e^{-\lambda t}$ or

$$N_s = N_0 e^{-\lambda t} \quad \text{--- eq. (10)}$$

Equation (10) is the equation of the graph of
 survivor against time (survivor curve). Similarly
 from eq. (1) it can be easily shown that

$$N_f = N_0 (1 - e^{-\lambda t}) \quad \text{--- eq. (11)}$$

$$\Rightarrow N_f \uparrow$$

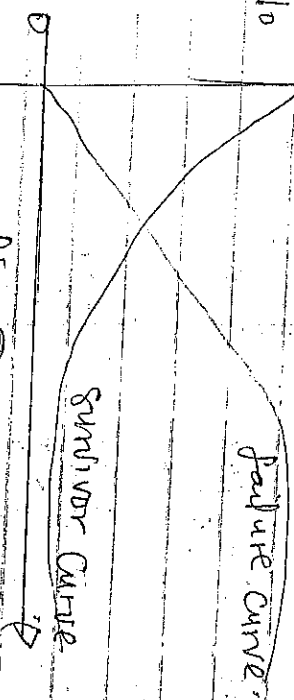


Fig. (2)

The fig. (2) above shows survivor and failure curve
 from eq. (1) $- R = \frac{N_s}{N_0} = e^{-\lambda t}$

Also from equation (3) $1 - \frac{N_f}{N_0} = (1 - e^{-\lambda t})$

EFFECT OF GENERATED HEAT

In fact semi conductors or devices like transistors that carry appreciable current ^{during} operating will most likely generate appreciable amount of heat. The heat generated can cause failure of such device which if exceeds the proper rating. It can also cause the variation of the normal increase in chemical reactions resulting in rapid ageing.

RELIABILITY

By derating a device made to operate at the level of the applied voltage. Current of power less likely to exceed excessive heat to be generated internally. Derating increases the temperature rise due to internal heat dissipated, thus decreases the failure rate.

INDICIAL EXERCISES

RELIABILITY PREDICTIONS

Reliability Prediction is the process of calculating the anticipated system reliability from assumed component failure rate. When calculating the reliability of systems which are made up of number of sub unit each having their own reliability the type of system must be defined the three major systems involved are

(i) Series System.

(ii) Parallel System.

(iii) The general guidelines/rules for the determination of

the system reliability involves the following steps:

(a) Identify the units in series within the system and

calculate the reliability of a single component unit

Using the relationship

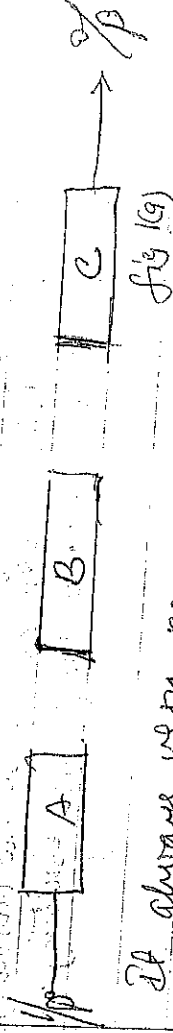
$R_s = R_1 + R_2 + \dots + R_n$

(b) Reduce each parallel arrangement to a single equivalent

Using eg. to get (b) (i) and (ii) respectively

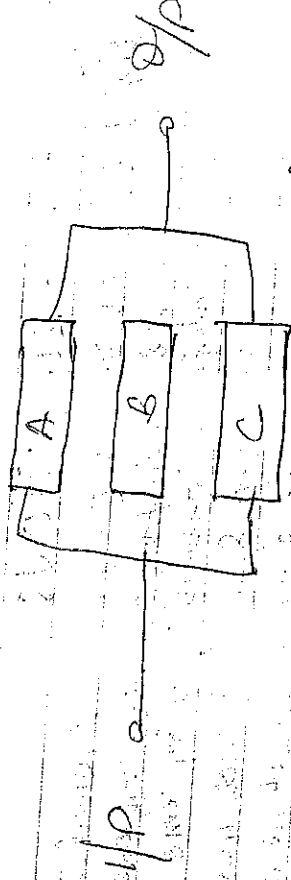
Find the product of the results in (i) and (ii) respectively

to get the final system reliability R_{sp}
 then $R_p = 1 - (1 - R_1)(1 - R_2) \dots (1 - R_n)$
 then $R_{sp} = R_p \times R_p = (R_1 R_2 \dots R_n) [1 - (1 - R_1)(1 - R_2) \dots (1 - R_n)]$
 SERIES SYSTEMS



It always very necessary to represent the system as number of functional blocks, inter connected according to the effect of each block failure on the total system block. Series system is a kind of system that if one of the blocks have failure i.e. if one block A fails to operate then B and C are all faulty and not functional.

PARALLEL SYSTEM



A parallel system arrangement is one which does not fail as a result of just one subunit that has gone faulty i.e. failure in one subunit or block can not cause failure of other subunit if block A becomes faulty block B and C will still function perfectly. Failure can only occur in a parallel arrangement if all the blocks A, B & C are all faulty.

SERIES PARALLEL SYSTEM

A series parallel system is a kind of system that fails if the blocks in series connection fail or if block ABC are all faulty as shown in fig 1(c) below.

we can conclude that a series connection the probability of a successful series system is given by

$$P_{abc} = P_a \times P_b \times P_c$$

Note also that the concept of multiplication rule of probabilities is useful in determining the reliability of series functions

ADDITIONAL

RULE

Referring to fig 1(b).

Let imagine there are two subunit A & B & C connected in parallel. If each of the sub unit can operate separately and

their individual probabilities of successful operations are known as P_a , P_b and P_c for subunit A, B & C

$$= P_a + P_b + P_c - P_a P_b P_c \quad \text{eq (3)}$$

Alternatively

$$P_{abc} = 1 - [(1 - P_a)(1 - P_b)(1 - P_c)] \quad \text{eq (4)}$$

Note that eq (3) and (4) are the same. Generally, $P_{abc \dots n} =$

$$1 - (1 - P_a)(1 - P_b)(1 - P_c) \dots (1 - P_n) \quad \text{eq (5)}$$

The expression equation (5) is known as the additional rule of probabilities is particularly useful in determining the parallel system respectively.

If R_s is the reliability of the series system where R_1, R_2

are the probability of the individual subunits

$R_s = R_1 \cdot R_2 \dots R_n$. If however the system is made up of exponentially failing units such that the sub-

unit reliability are $\lambda_1, \lambda_2, \dots, \lambda_n$ at

$$R_1 = e^{-\lambda_1 t} \quad R_2 = e^{-\lambda_2 t} \quad R_n = e^{-\lambda_n t}$$

λ_1, λ_2 and λ_n are the respective failure rate of subunit then

$$R_s = R_1 \cdot R_2 \dots R_n = e^{-\lambda_1 t} \times e^{-\lambda_2 t} \dots e^{-\lambda_n t}$$

$$= e^{-(\lambda_1 + \lambda_2 + \dots + \lambda_n)t}$$

$$= e^{-\lambda_s t}$$

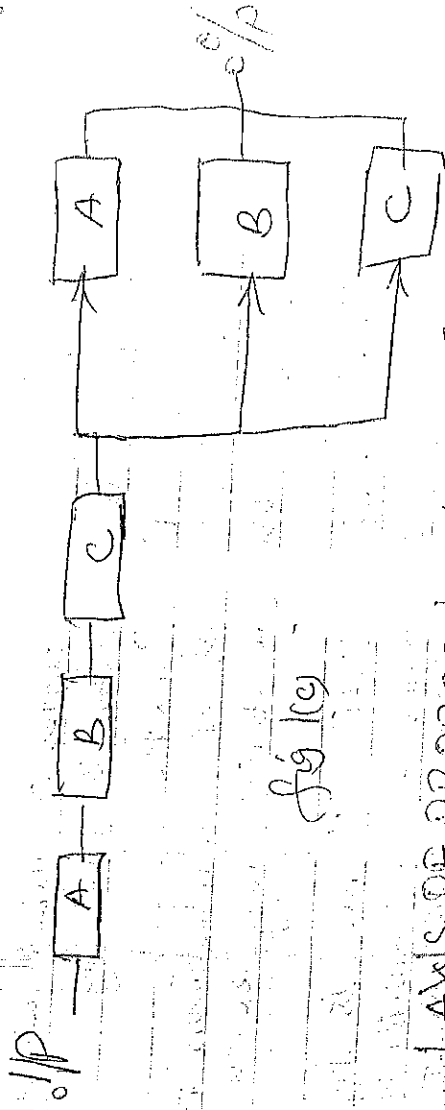


Fig 1(c)

LAWS OF PROBABILITY RELEVANT RELIABILITY PREDICTIONS

Recall that reliability means successful operation. Calculating system reliability various laws in probability theory are used. Below are the relevant laws of probability used in predicting reliability systems. Multiplication Rule (LAW) Referring to fig 1(c) above, and imagine the two subunit connected in series with both subunit can operate simultaneously and the individual probability operating successes are shown. or knowing, then the probability of successful operation of the series system is the product of the individual probabilities.

$P_{abc} = P_a \times P_b \times P_c \dots P_n$ eq (1)
where P_a is equal to the probability that A will operate successfully while P_b is the probability that B will operate successfully and P_c is the probability that C will operate successfully.

Considering A, B, C n^{th} term connected in series then the probability of successful operation n will be $P_{abcn} = P_a \times P_b \times P_c \dots P_n$ eq (2)
The above equation is generally referred to as the multiplication rule of the probability eq.
 $P_a = 0.9$ $P_b = 0.8$ $P_c = 0.65$ Calculate the system reliability
Sol.

$$P_{abc} = P_a \times P_b \times P_c$$

$$= 0.9 \times 0.8 \times 0.65 = 0.48$$

where λ_s is the system failure rate given by λ_s
 $= \lambda_1 + \lambda_2 + \dots + \lambda_n$ eq (1)

RELIABILITY OF A SERIES SYSTEM
 The mtr of a series system $MS = 1$

$$MS = \frac{1}{\lambda_s} = \frac{1}{\lambda_1 + \lambda_2 + \dots + \lambda_n} \quad \text{eq (2)}$$

For a system in series combining similar subunit of equal reliability the system reliability $R_s = e^{-\lambda_s t}$
 $\Rightarrow$ System failure rate $\lambda_s t = n \lambda t$

$\therefore$ System mtr of $MS = \frac{1}{n}$

RELIABILITY OF A PARALLEL SYSTEM

In line with equation (5) the reliability of a parallel system with different subunit

$$R_p = 1 - [(1-R_1)(1-R_2)(1-R_3) \dots (1-R_n)] \quad \text{eq (3)}$$

$R_1, R_2, \dots, R_n$ are the reliability of the respective subunit. Note that also if

$$R_1 = R_2 = R_n = R \quad \text{therefore}$$

$R_p = 1 - [1 - (1-R)^n]$ if the system is of the exponentially failing subunit such that

$$R_1 = e^{-\lambda_1 t}, R = e^{-\lambda t} \quad \dots \quad R_n = e^{-\lambda t} \quad \text{therefore}$$

System reliability of a system with n th subunit is given by

$$R_p = 1 - [1 - (1 - e^{-\lambda t})^n] \quad \dots \quad (1 - e^{-\lambda t})^n$$

(for a system with two subunits in parallel equals

$$R_{p2} = 1 - (1 - R_1)(1 - R_2) = R_1 + R_2 - R_1 R_2$$

$$= e^{-\lambda_1 t} + e^{-\lambda_2 t} - e^{-(\lambda_1 + \lambda_2)t}$$

SYSTEM WITH THREE SUBUNIT
 $R_p = 1 - (1 - R_1)(1 - R_2)(1 - R_3)$

$$= 1 - (1 - R)^3$$

Expanding and simplifying the expression on the RHS we have

$$R_P = R_1 + R_2 + R_3 - R_1 R_2 R_3 - R_1 R_2 - R_1 R_3 - R_2 R_3$$

therefore

$$= \lambda_1 t - \lambda_2 t - \lambda_3 t - (\lambda_1 + \lambda_2) t - (\lambda_2 + \lambda_3) t - e^{-\lambda_1 t} - e^{-\lambda_2 t} - e^{-\lambda_3 t} - e^{-(\lambda_1 + \lambda_2 + \lambda_3) t}$$

MTBF OF A PARALLEL SYSTEM

We recall the MTBF can be expressed as the integral reliability with the limit of integration from

$$M = \int_0^{\infty} R(t) dt$$

The MTBF can be obtained by integration of the reliability expression, thus method of finding the MTBF does not rely on the expression being of the exponential form. Consequently the reliability of parallel system with two subunit system is given by

$$MP_2 = \int_0^{\infty} R_P dt = \int_0^{\infty} (R_1 + R_2 - R_1 \times R_2) dt$$

$$MP_2 = \frac{1}{\lambda_1} + \frac{1}{\lambda_2} - \frac{1}{\lambda_1 + \lambda_2}$$

for three units

$$MP_3 = \frac{1}{\lambda_1} + \frac{1}{\lambda_2} + \frac{1}{\lambda_3} - \frac{1}{\lambda_1 + \lambda_2} - \frac{1}{\lambda_1 + \lambda_3} - \frac{1}{\lambda_2 + \lambda_3} + \frac{1}{\lambda_1 + \lambda_2 + \lambda_3}$$

for n units

$$MP_n = \frac{1}{\lambda_1} + \frac{1}{\lambda_2} + \frac{1}{\lambda_3} + \dots + \frac{1}{\lambda_n} - \dots - \frac{1}{\lambda_1 + \lambda_2 + \dots + \lambda_n} \quad \text{--- eq. (9)}$$

Exercise

Compare the reliability of a series system with parallel system if each system contains three subunits having reliability of 0.95, 0.85 and 0.75 respectively.

Solution

(i) Series Solution

For a combination of three subunit the reliability $R_S = 0.95 \times 0.85 \times 0.75 = 0.605$

(ii) For the parallel arrangement

$$R_P = [1 - (1 - 0.95)(1 - 0.85)(1 - 0.75)]$$

$$R_P = [1 - (0.05 \times 0.15 \times 0.25)] =$$

$$R_P = [1 - 0.001875]$$

$$R_P = 0.998125$$

(ii)

A computer and two other computer are available for construct of a channel such that if a computer fails another is simultaneously switched for operation. The failure rate of each computer is 0.01 failure per hour. Calculate the system reliability if the operating period is 50 hrs and the switch is perfect.

Solution

One standby computer

Two stand by computer.

For a single computer from equation 8 we have $n=1$

$$R(t) = e^{-\lambda t}$$

$$= e^{-(0.01)(50)}$$

$$= e$$

$$= e^{-(0.5)}$$

$$= e$$

$$= 0.606$$

(ii) For One Standby Computer $n=2$

$$R_{2t} = e^{-\lambda t} (1 + \lambda t)$$

$$= e^{0.5} (1 + 0.5)$$

(iii) For Two Standby Computer $n=3$

$$R_{3t} = e^{-\lambda t} \left(1 + \lambda t + \frac{\lambda^2 t^2}{2!} \right)$$

$$= e^{-0.5} (1 + 0.5 + 0.125) = 0.985$$

Q8. A complex Communication Satellite has an inbuilt microwave repeater unit having mean time to failure of 40,000 hours. The line is operative if one channel is working and the reliability of the switching unit is 0.75. Calculate the

(i) Single Channel Reliability

(ii) Reliability of the 2 Parallel Channels

(iii) The Reliability of the three Parallel channels

(iv) Solution.

For Single Channel



Two Pc



2Pc



3Pc



O/p

From our data given above,

Failure Rate $\lambda = \frac{1}{4} \times 10^4$

$$= 0.25 \times 10^{-4} / h$$

Operating period $t = 1 yr = 24 \times 365 = 8760 h$
 $\therefore$ single channel reliability $= R_1 = R = e^{-0.25 \times 8760 \times 10^{-4}}$

$$R_1 = \underline{\underline{0.803}}$$

For 2 channels in parallel
 Reliability $R_2 = 1 - (1 - R_1)(1 - R_1)$

$$\begin{aligned} &= 2R - R^2 \\ &= 2(0.803) - (0.803)^2 \\ &= 1.606 - 0.644809 \\ &= \underline{\underline{0.961}} \end{aligned}$$

Considering the presence of its switching system which also parallel system combination so that

$$\begin{aligned} R_2 &= R_2 \times R_3 = 0.961 \times 0.95 \\ &= \underline{\underline{0.913}} \end{aligned}$$

For 3 channels.

$$\begin{aligned} \text{Reliability } R_3 &= 1 - (1 - 0.803)^3 \text{ where } R_1 = 0.803 \\ R_3 &= [1 - (1 - R_1)^3] \end{aligned}$$

$$R_3 = (1 - 0.803)^3 = \underline{\underline{0.992}}$$

$$\text{Similarly, } R_{3S} = R_3 \times R_2 = \underline{\underline{0.992}}$$

$$\therefore R_{3S} = 0.992 \times 0.95 = \underline{\underline{0.942}}$$

844
 800 Components were tested for the period testing for 5000 h
 If 150 Components survived without failure. Assuming wear out failure, the mean time to failure of the component are recorded in the table below.

Sl. No	N ^o of Components	Time to failure in hrs
1	10	400
2	8	5000
3	7	800
4	5	900

Calculate; total test hours before failure

(i) Total test hours without failure

(ii) Total survival hours

(iii) MTBF

Solution

(i) Total test hours before failure

$$= (10 \times 400) + (8 \times 500) + (7 \times 800) + (5 \times 900)$$

$$= 4000 + 4000 + 5600 + 4500$$

$$= 18100 \text{ hrs.}$$

(ii) Total test hours without failure

$$(500 \text{ hrs} \times 150 \text{ Components}) = 75000 \text{ hrs}$$

(iii) Total survival hours

$$75000 \text{ hrs.}$$

$$+ 18100 \text{ hrs}$$

$$= 93100 \text{ hrs}$$

$$\text{iv. MTBF (M)} = \frac{\text{Total survival test hours}}{\text{No. of components}}$$

$$= \frac{93100}{50}$$

$$= 1862 \text{ hrs}$$

REDUNDANCY

Redundancy in reliability is defined as the provision of more than one means of getting an equipment/system to perform a given function or perform a certain task. The application of redundancy is therefore said to be a method of enhancing the equipment/system reliability. This implies that by the application of redundancy on equipments/systems they will continue to function satisfactorily despite failure of some items from which it has been built up from.

SIGNIFICANCE OF REDUNDANCY

eg. The Pilot in an aircraft needs an accuracy for allimeters to indicate how high he/she is flying. Since the Pilot regards this information vital, provision has to be made for the situation where there is a fault. One way out is to provide these allimeters in air craft so that if one goes faulty any two gives the same reading, ^{identical}

In this example it is clear that one allimeter may be enough for the flight's measurement but provision is made for additional two redundant allimeters in order to achieve system reliability

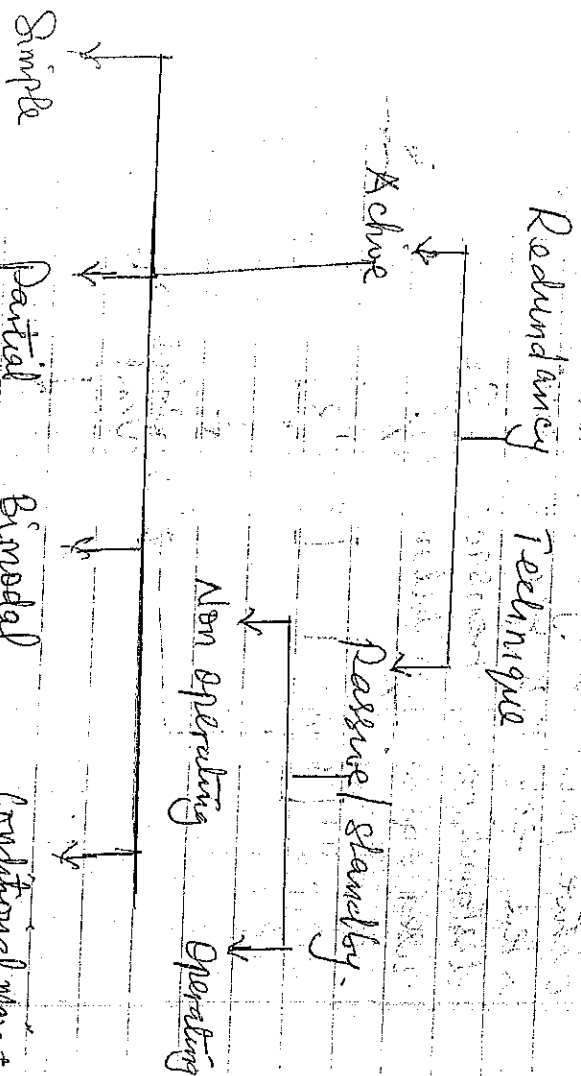
LIMITATIONS

The application of redundancy however is not without limitation. It increases weight, space competition, cost and time to design and maintain.

CLASSIFICATION OF REDUNDANCY

Achieve Redundancy Standby Redundancy (Passive)

The two types of Redundancy are further classified into different headings as shown below

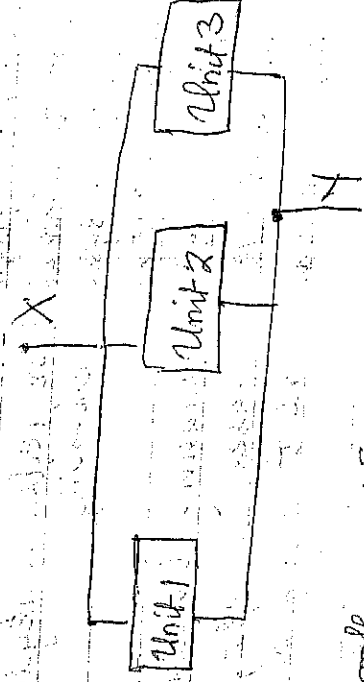


It is to be noted that certain types of redundancy

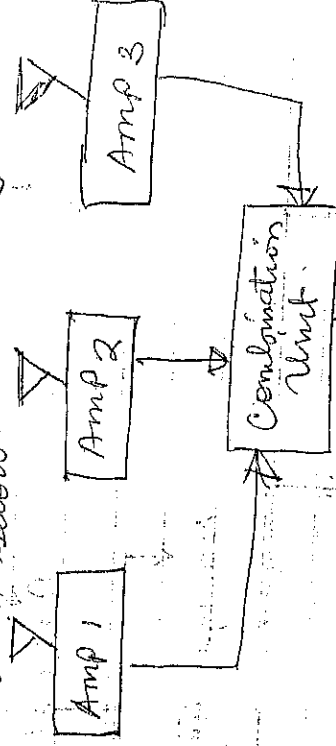
applicable ~~part~~ at the part of circuit level that at the equipment / system for the purpose of this study we will consider the two major types of redundancy.

ACTIVE REDUNDANCY

Under the condition of active redundancy all the units are energized simultaneously to perform a given task or functions. The diagram below shows three additional units.



The working noting that the units are drawn through key are in parallel in the electrical sense. The electrical analogy is that current will flow X - Y so long as there is an unbroken circuit through at least one unit. An application ~~this example~~ of the application of active redundancy is further illustrated below. This is the technique of a diversity reception in the long distance radio transmission. Generally space diversity is used as shown below.



TO Processing Unit
DIVERSITY RECEPTION IN LONG DISTANCE
RADIO TRANSMISSION LINES

After signal amplification the combination automatically select the largest of the three signal received by the aerial for subsequent processing. In this case all the three aerials are actively working at the same time.

MATHEMATICAL BASES FOR ACTIVE FULL REDUNDANCY

In line with the concept of active redundancy in this case a number of units are employed simultaneously but system operation is maintained so long as at least one unit remains operating. Under this condition additional rule applies because the reliability of such a system is the probability of non failure one or more of the units. Hence in terms that are in parallel arrangement

$$R_P = [1 - (1 - R_1)(1 - R_2)(1 - R_n)]$$

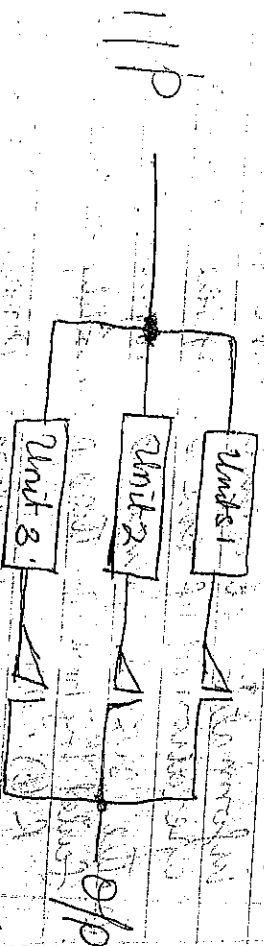
The above for unit system is

$$MP = \frac{1}{\lambda} + \frac{1}{\lambda} + \frac{1}{\lambda} + \dots + \frac{1}{\lambda}$$

where λ is equal to failure rate of each unit

STAND BY REDUNDANCY

Standby redundancy involves the use of additional units which are only engaged when the operating unit fails. The figure below shows n identical units, one operating when a failure is detected then the unit will switch in its place



Standby Redundancy

CLASSIFICATION OF AVAILABILITY

Availability is an important metric used to assess the performance of repairable system accounting for both reliability and maintainability properties of component or systems.

The classification of availability is somewhat flexible and is largely based on the type down times used in computation and on the relationship with time (i.e. the span of time to which the availability refers). As a result the number of classification of availability includes:

(i) Instantaneous (or Point availability)

(ii) Steady State availability

(iii) Inherent availability

(iv) Achieved availability

(v) Operational availability

(vi) Average uptime availability (or Mean availability)

INSTANTANEOUS (OR POINT AVAILABILITY) Instantaneous or point availability is the probability that a system or component will be operational (up and running) at a specific time t . This classification is typically used in the military, as it is sometimes necessary to estimate the availability of a system at a specific time of interest i.e. when a certain mission is to be completed. The point availability is very similar to the reliability function in that it gives a probability that a system will function at the given time t . Unlike reliability however the point

availability measures incorporates maintanability information at a given time t , the system will be operational if one or more conditions is satisfied.

The system functioned properly from 0 to t , i.e. it never failed by time t . The probability of this happening is $R(t)$, or

(ii) The system functioned properly since the last repair at time μ , $0 < \mu < t$. The probability of this condition is

$\int R(t - t) m(t) dt$ with $m(t)$ being the renewal density function of the system consequently the point availability is summation of the above two probabilities or

$$A(t) = R(t) + \int R(t - u) m(u) du.$$

2. STEADY STATE AVAILABILITY: The steady state availability of the system is the limit of the availability function as time t tends to infinity. Steady state availability is also called long run or asymptotic availability. A common equation for the steady state availability as found in literature is $A(\infty) = \lim_{t \rightarrow \infty} A(t)$. Note that steady state availability also applies to mean availability. The figure below illustrates steady state availability.

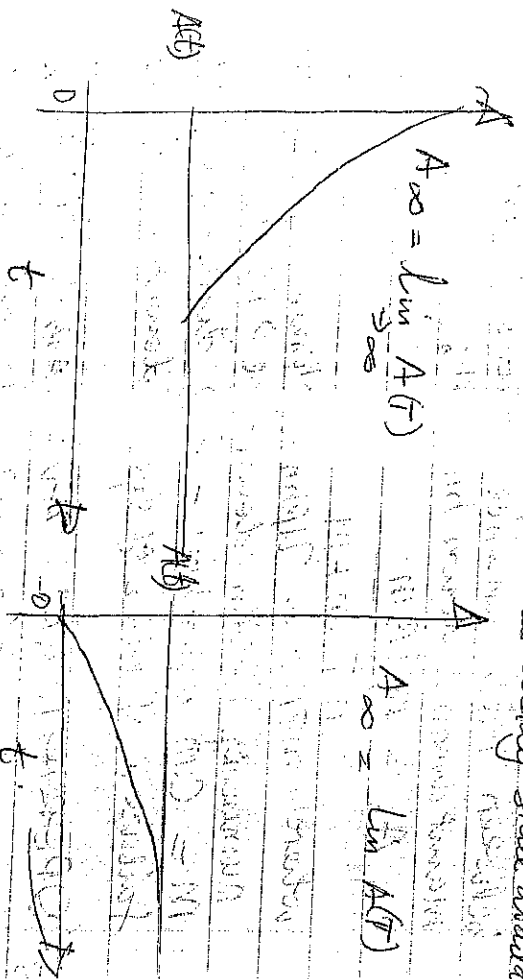


fig 1(a)

fig 1(b)

The figures above shows the illustration of point availability approaching steady state.

3. MAINTENANCE AVAILABILITY: This availability is concerned only with corrective maintenance (CM) down time of the system. This classification is what sometimes referred to as the availability seen by the maintenance personnel. For a single component, the inherent availability can be computed by

$$AI = \frac{MTTC}{MTTC + MTTR} \text{ for computer systems we have}$$

$$A_I = \frac{MTBF}{MTBF + MTTR}$$

$$\text{where } MTBF = \frac{\text{Uptime}}{\text{Number of system failure}}$$

$$MTTR = \frac{\text{Downtime}}{\text{Number of system failure}}$$

4. ACHIEVED AVAILABILITY: This is the steady state availability when considering corrective and preventive down time of the system. The achieved availability is sometimes referred to as the availability seen by the maintenance department including both corrective and preventive maintenance but does not include logistic delays, supply delay or administration delays. This can be computed by looking at the mean time between maintenance actions (MTBM): and the mean maintenance down time, M ;

$$A_A = \frac{MTBM}{MTBM + M}$$

$$\text{where } MTBM = \frac{\text{Uptime}}{\text{Number of system failure} + \text{Number of system downtime PMS}}$$

$$M = \frac{\text{CM downtime} + \text{PM downtime}}{\text{Number of system failure} + \text{Number of system downtime PMS}}$$

$$M = \frac{\text{CM downtime} + \text{PM downtime}}{\text{Number of system failure} + \text{Number of system downtime PMS}}$$

5. OPERATIONAL AVAILABILITY: This is a measure of the real average availability over a period of time and includes all experienced sources of downtime, such as administrative downtime logistic downtime etc.

This is the key type of availability that the customer actually experiences. It is essentially posterior availability based on actual events that happened to the system.

$$A_o = \frac{\text{Uptime}}{\text{Operating System}}$$

where the operating system is the overall time period of the operation being investigated and uptime is the total time the system was functioning during the operating cycle.

AVERAGE UPTIME OR MEAN AVAILABILITY ⁽⁴⁾

The mean availability is the proportion of time during mission or time period that the system is available for use. It represents mean value of the instantaneous availability function over the period (0, T) and is given by

$$\overline{A(t)} = \frac{1}{T} \int_0^T A(t) dt$$

For systems that have periodic maintenance availability may be zero at regular periodic intervals. In this case mean availability is a more meaningful measure than instantaneous availability. Such definition of availability is commonly used in manufacturing and telecommunication systems.

DEFINITION OF TERMS

1. Down time:- This refers to the period of time when a machine or factory is not in operating mode especially as a result of faults in the system.
2. Up time:- This refers to a period of time when a machine or factory is operating and available for use to accomplish specific tasks.
3. Maintainability:- This is defined as the probability that a system or system element can be repaired in a defined function under stated condition without failure for a given period of time. A precise definition must include a detailed description of the function, the environment, the time scale and what constitutes a failure.
4. Availability:- This is defined as the probability that a repairable system or system element is operational at a given time under a given set of environmental conditions. Availability depends on reliability and maintainability.
5. Fault:- The event(s) or inoperable state in which any

Item or part of an item does not or would not perform as expected.

6. Unavailability:- This is the probability that an item will not operate correctly at a given time and under specific environmental or physical conditions. It opposes availability numerical values associated with the calculation of availability which are often awkward consisting with series of 96 before reaching any significant numerical information for the reason it is more convenient to use the complement measure of availability namely Unavailability. Expressed mathematically Unavailability is 1 minus the availability. Therefore a system with availability 0.99999665 is more concisely described as having an Unavailability of 3.4668×10^{-6} . Unavailability may be expressed mathematically as the ratio

$$U = \text{MTTR} / \text{MTTF}$$
 where MTTR is the mean time to repair and MTTF is the mean time to failure. Alternatively this can be written as

$$U = 1/M$$
 where M = Critical failure rate and M is the critical repair rate. In telecommunication an unavailability is an expression of the degree to which a system, subsystem or equipment is not operational and not in a commutable state at the start of mission where the mission call for an unknown time is (random).

7. Maintainance Actions These are those leadership actions taken by one or more members of a group to enhance the social relationship among group members. This action's turn to increase the overall effectiveness of the group and create a more positive atmosphere of interaction with the group. Maintainance actions contrasted with task actions which are those actions taken to enable the group to complete a specific task.

8. Derating:- In electronics derating is the operation of a device at less than its rated maximum capability in order to prolong the maximum power rating.